OFFICIAL ARCHITECTURE & USER GUIDE v2.5 Production Edition



# PromptVault Pro

The Comprehensive Interactive Guide & Technical Reference for Managing, Parameterizing, and Executing Production-Grade AI Prompts & Blueprints.

- ⚙️ Automation & Systems
- 💻 Coding & Software Architecture
- 💼 Business & Product Strategy
- 📈 Marketing, Content & SEO
- 🎨 Image & Multimedia Generation
- ✨ Design, UI/UX & Web Dev
- 🎓 Education & Skill Building

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**Repository:** github.com/rockyswartz/PromptLibrary
**Web App:** rockyswartz.github.io/PromptLibrary

**Format:** Fully Interactive Print-Ready PDF
**Includes:** Process Infographic, UI Element Maps, FAQs & Clickable Index

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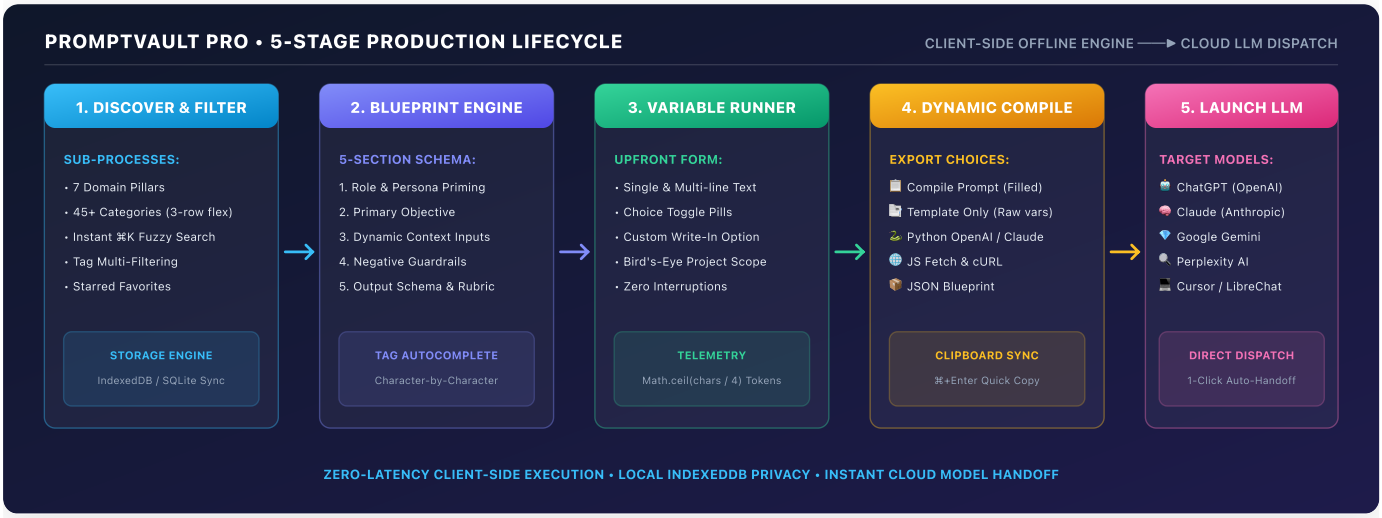
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# 📊 Architecture & End-to-End Workflow Infographic

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A comprehensive architectural diagram mapping user progression from taxonomy discovery through upfront parameterization, compilation, and 1-click cloud LLM execution.



# 1. System Architecture & Main Workspace Tour

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PromptVault Pro provides a unified, local-first workspace for finding, customizing, parameterizing, and executing enterprise-grade AI blueprints. All private prompt data stays entirely on your local machine.

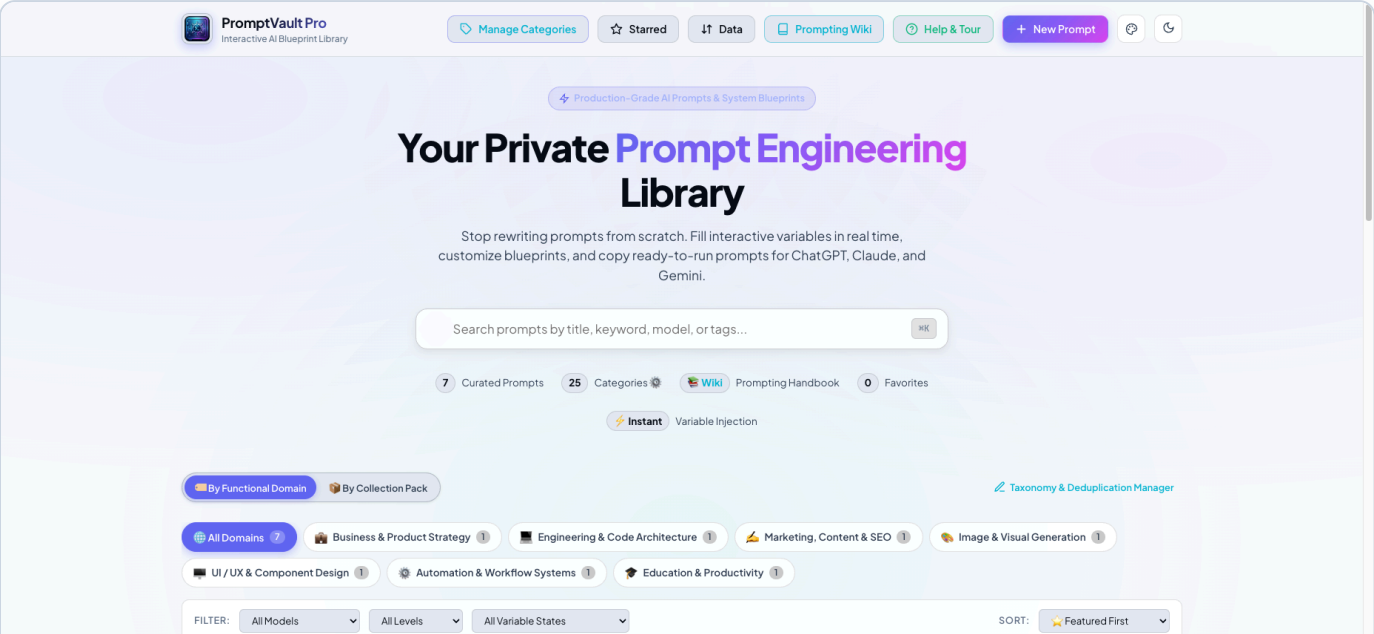



**Figure 1.1:** PromptVault Pro Main Workspace, 7 Architectural Pillars & Production Prompt Card Grid **Live Workspace View**

## Main Workspace UI Element Reference

| UI ELEMENT | COMPONENT LOCATION | FUNCTIONAL DESCRIPTION & BEST PRACTICE |
|---|---|---|
| **1. 7 Domain Pillars** | Top pill tabs below hero header | Filters your library across 7 engineering domains: Automation, Coding, Business, Marketing, Multimedia, Design, and Education. Enforces clean cognitive alignment. |
| **2. Global Action Bar** | Top-right navigation bar | Features quick access buttons: `Manage Categories` (Taxonomy), `Starred` (Favorites), `⇅ Data` (Hub for import/export), `Prompting Wiki`, `Help & Tour`, and `+ New Prompt`. |
| **3. Instant Fuzzy Search** | Centered hero search input ( `⌘K` ) | Real-time substring search across titles, descriptions, modular sections, parameter tags, authors, and hashtags with zero latency. |
| **4. 3-Row Category Limit** | Category flex pill bar | Limits category tags to 3 horizontal rows maximum. Additional categories are neatly bundled into a reactive `+ More Categories (N)` pill button to prevent navigation sprawl. |
| **5. Prompt Cards & Badges** | Main responsive card grid | Every prompt card displays its Domain Pillar badge, Category tag, Blueprint indicator, Difficulty, dynamic variable highlights, author credit, and 1-click `⚡ Use Prompt` runner trigger. |

## The 7 Core Architectural Domain Pillars

PromptVault Pro categorizes knowledge across 7 immutable domain disciplines:

- ⚙️ **Automation & Systems:** Robotic process automation, spreadsheet formulas, webhooks, API pipelines.
- 💻 **Coding & Architecture:** Full-stack development, refactoring, algorithms, testing, system architecture.
- 💼 **Business & Strategy:** Pitch decks, executive memos, competitive analysis, PRDs, strategic roadmaps.
- 📈 **Marketing & SEO:** High-conversion copywriting, technical SEO audits, social campaigns, email sequences.
- 🎨 **Image & Multimedia:** Midjourney, DALL-E, Stable Diffusion parameter specs, cinematography direction.
- ✨ **UI/UX & Web Dev:** Responsive component specs, design systems, accessibility (a11y), Tailwind/CSS architecture.
- 🎓 **Education & Skill Building:** Socratic tutoring, concept deconstruction, complex topic explanations.

# 2. The 3-Tier Taxonomy & Deduplication Shield

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To prevent prompt libraries from devolving into chaotic, duplicate-ridden tag clouds, PromptVault Pro implements a rigid 3-Tier Taxonomy Tree backed by an interactive Deduplication Manager.

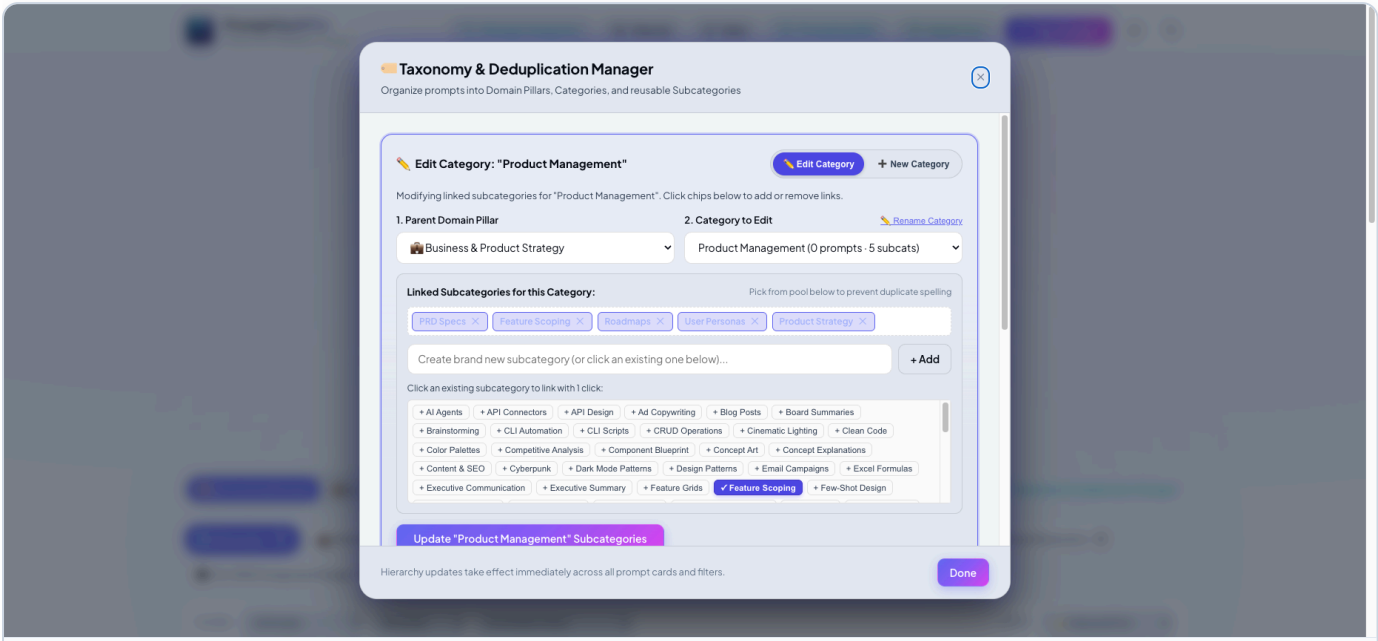



**Figure 2.1:** Taxonomy & Deduplication Manager Modal with Mode Switcher and Master Subcategory Pool — **Taxonomy Manager View**

## Taxonomy Manager UI Element Reference

| UI ELEMENT | CONTROL TYPE | FUNCTIONAL DESCRIPTION & BEST PRACTICE |
|---|---|---|
| **1. Domain Switcher** | Pill tab bar across modal top | Switches the active Domain Pillar (e.g. Business, Automation, Coding) to manage its categories and nested subcategories. |
| **2. Mode Switcher** | `[ ✏️ Edit Category ]` vs `[ + New Category ]` | Toggle between modifying an existing category's subcategories and creating a new category from scratch. Prevents competing dropdown conflicts. |
| **3. Category Dropdown** | Dropdown with telemetry counts | Displays categories along with real-time stats: e.g. `Executive Presentations (3 prompts, 9 subcats)`. Selecting a category instantly pulls its subcategories into view. |
| **4. Inline Rename Button** | `✏️ Rename Category` action button | Allows renaming any category in place. Automatically executes a cascading database update across all assigned prompts so no cards are orphaned. |
| **5. Master Subcategory Pool** | 100+ Interactive Chip Matrix | Lists all available subcategories. Active/linked subcategories are highlighted with **vibrant solid accent backgrounds (** `✓ Roadmaps` **)**; unlinked appear neutral ( `+ Name` ). 1-click toggles link state. |
| **6. Active Hierarchy Accordion** | Collapsible domain accordion | Shows the full category-to-subcategory tree for the active pillar. Each card includes a direct ✏️ **Edit** button and a safe delete button (prompts reassign to *General*). |

# 3. Prompt Authoring, 5-Section Blueprints & Tag Autocomplete

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PromptVault Pro supports both rapid single-text prompts and enterprise 5-section blueprints designed for high-reliability outputs.

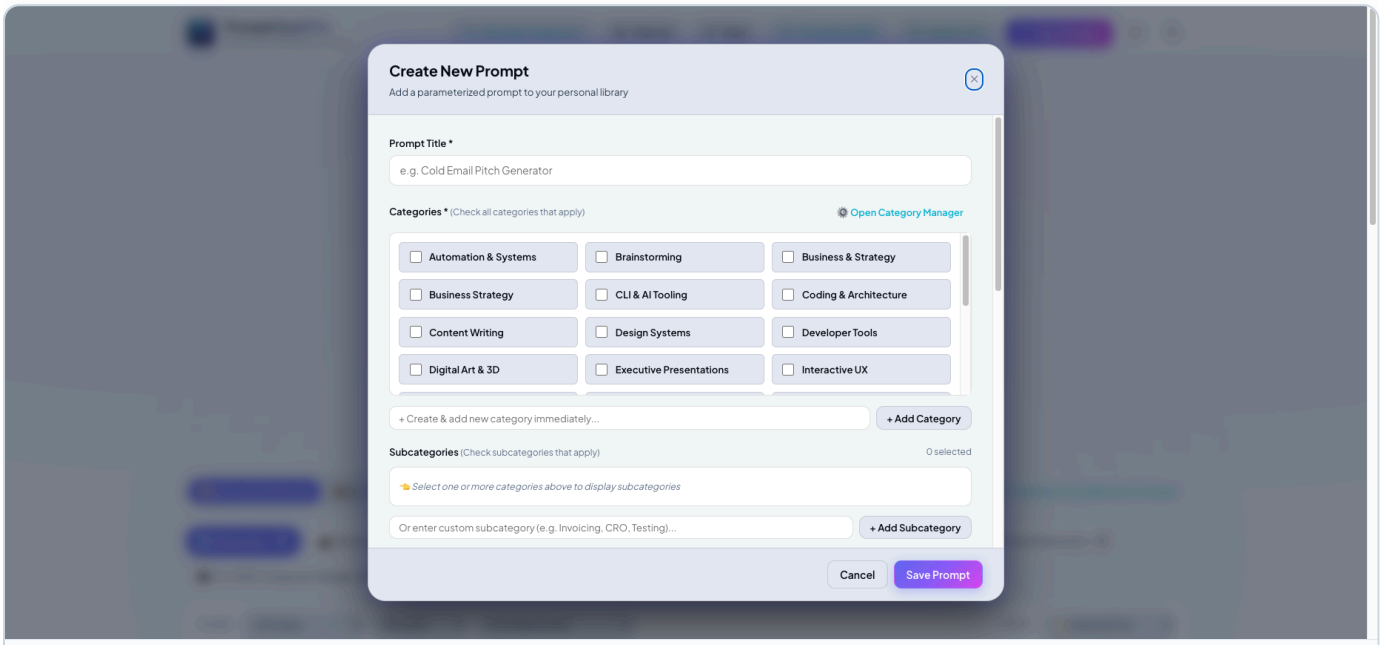



**Figure 3.1:** Create / Edit Prompt Modal with Blueprint Sections, Tag Autocomplete & Originator Credits **Prompt Form View**

## Prompt Authoring UI Element Reference

| UI ELEMENT | FORM SECTION | FUNCTIONAL DESCRIPTION & BEST PRACTICE |
|---|---|---|
| **1. Title & Difficulty** | Header section | Assign a descriptive action title and cognitive level (Beginner, Intermediate, Advanced, Expert) to guide users. |
| **2. Multi-Category Selector** | Category Checkbox Grid | Prompts can belong to multiple categories simultaneously. Selecting a category surfaces its contextual subcategory tags for 1-click addition. |
| **3. 5-Section Blueprint** | Structured Blueprint Editor | Separates Persona, Primary Objective, Dynamic Context Inputs (with `{variables}`), Negative Guardrails, and Output Rubric. |
| **4. Tag Autocomplete** | Tag input field | WordPress-style live autocomplete dropdown. As you type (e.g. `s` then `o`), shows matching tags with usage counts: `seo (12)`. Keyboard navigable with arrow keys. |
| **5. Originator Attribution** | Author & Author URL fields | Gives visible credit on prompt cards to creators (e.g. 👤 Randy Austin - ExcelForFreelancers, 👤 Anthropic, 👤 HubSpot). |
| **6. Safe Delete Button** | Bottom-left red action button | Deletes the prompt with a confirmation safety modal to prevent accidental data loss. |

# 4. Dynamic Variables & Live Multi-LLM Runner

↑ Top

Static prompts require manual find-and-replace editing. PromptVault Pro turns prompts into interactive software tools through dynamic variable parsing and upfront parameter pre-filling.

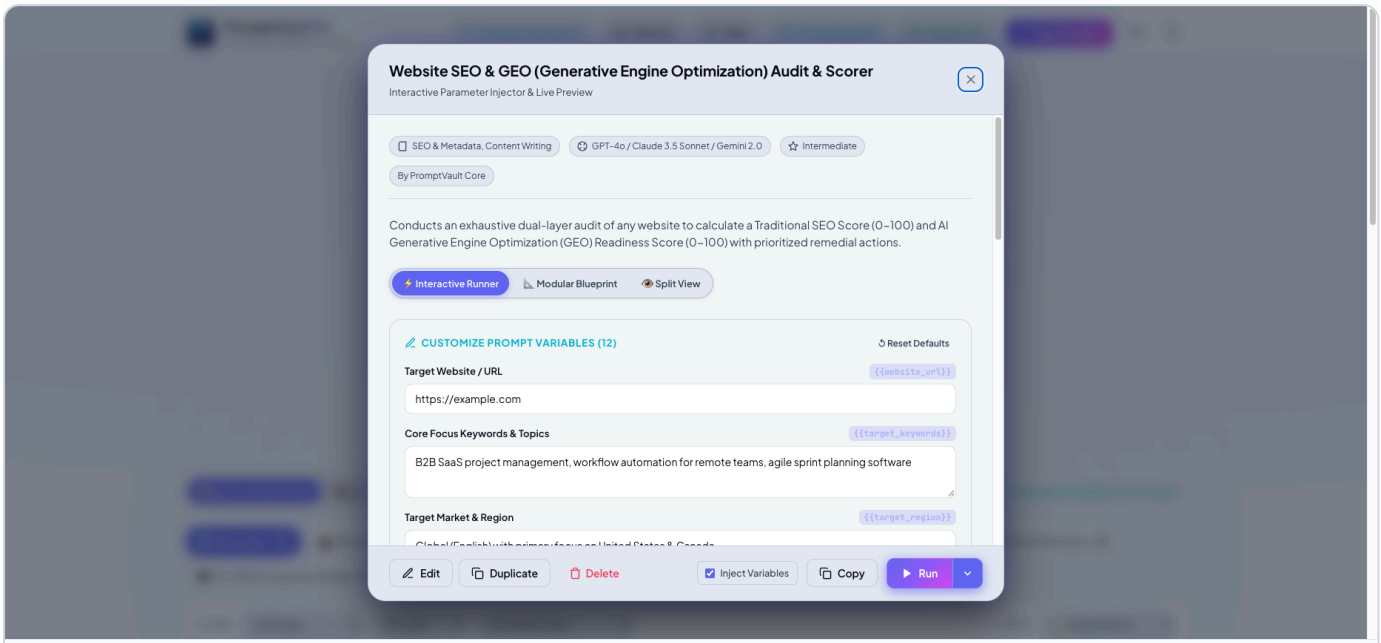



**Figure 4.1:** Live Variable Runner with Upfront Parameter Pre-fill, Choice Pills & Token Telemetry — Live Runner View

## Live Runner UI Element Reference

| UI ELEMENT | RUNNER COMPONENT | FUNCTIONAL DESCRIPTION & BEST PRACTICE |
|---|---|---|
| **1. Upfront Pre-fill Panel** | Top of runner modal | Consolidates all prompt variables into a single upfront form. Gives users a bird's-eye view of project scope and requirements before copying. |
| **2. Choice Toggle Pills** | Multiple-choice inputs | Renders choices defined in `{name: Opt 1 \| Opt 2}` as 1-click selectable pills, with a custom write-in fallback for bespoke options. |
| **3. Live Token Telemetry** | Telemetry counter bar | Calculates characters, words, and estimated LLM tokens (`Math.ceil(characters / 4)`) in real time as variables are adjusted. |
| **4. Compile vs Template** | Clipboard action buttons | Choose between copying the compiled prompt (with variables filled in) or copying the raw template (with placeholders intact for prompt engineers). |
| **5. 1-Click LLM Launcher** | `Run in LLM / Export ▾` button | Opens the Multi-LLM modal to push the compiled prompt directly to your preferred AI model in a new tab. |

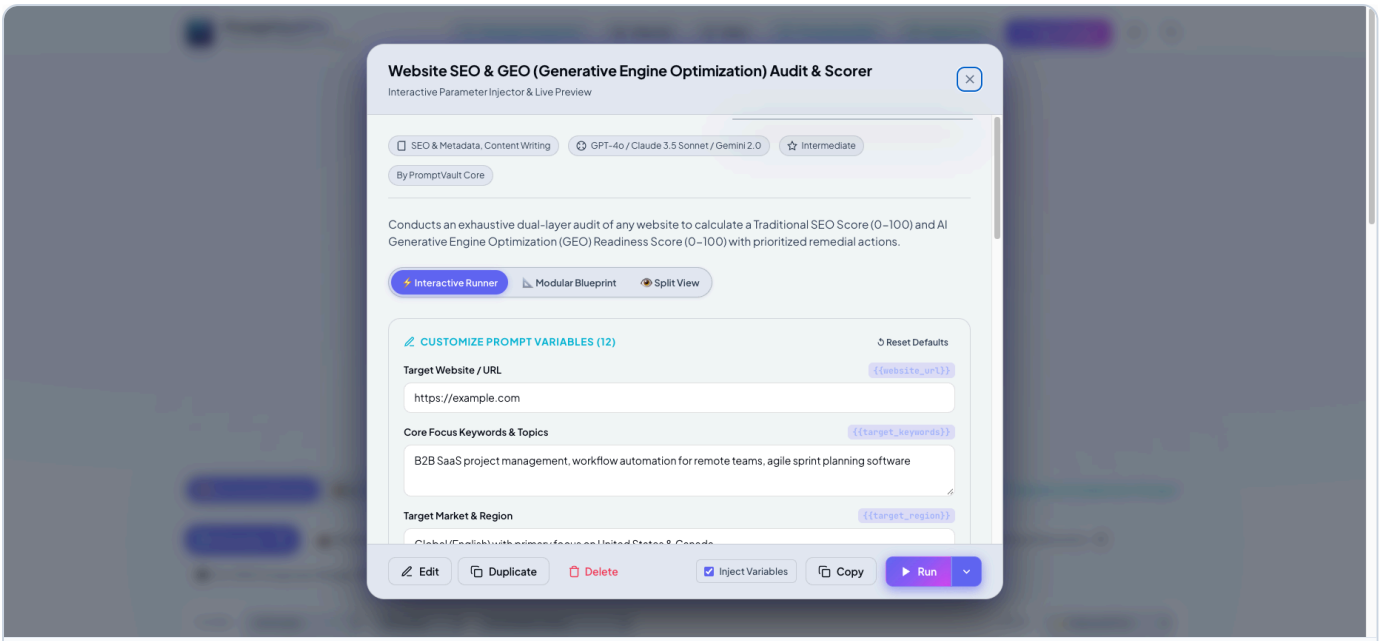



**Figure 4.2:** 1-Click Multi-LLM Launcher Popover (ChatGPT, Claude, Gemini, Perplexity, Cursor, LibreChat)

**LLM Launcher Popover**

# 5. ↑↓ Data Hub: Ingest, Backups & Turnkey Enrichment

↑ Top

The **↑↓ Data** Hub provides comprehensive data sovereignty, rapid text ingestion, and multi-format backup tools.

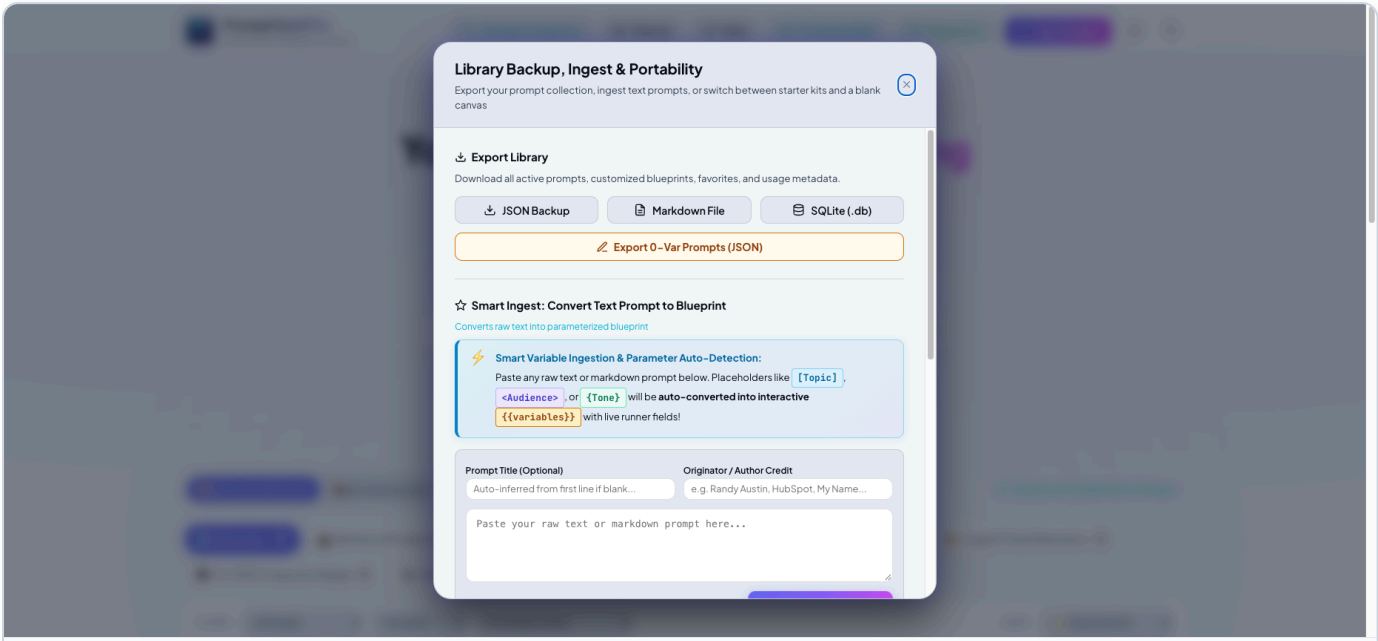



**Figure 5.1:** Data Management Hub with Canvas Presets, Smart Ingest, and Multi-Format Exports **Data Hub View**

## Data Hub UI Element Reference

| UI ELEMENT | ACTION TYPE | FUNCTIONAL DESCRIPTION & BEST PRACTICE |
|---|---|---|
| **1. Blank Canvas Preset** | `Start from Scratch` | Clears all prompts to let you build your own library, while preserving all 44 categories and 7 domain pillars. |
| **2. Starter Pack Preset** | `⚡ Load Starter Pack` | Restores the 7 flagship production blueprints across all domain disciplines. |
| **3. Smart Ingestion** | Raw text-to-prompt engine | Paste raw prompts with brackets ( `[Topic]` , `<Audience>` , `{Tone}` ). Automatically converts them into PromptVault variables, infers titles, and adds them to your library. |
| **4. Multi-Format Backups** | Export & Import actions | Download portable JSON backups, download the local SQLite `promptlibrary.db` file, or export documentation in Markdown. |
| **5. Turnkey LLM Prompt** | Copyable master prompt | A pre-engineered prompt to batch-parameterize exported 0-variable prompts using ChatGPT, Claude, or Gemini. |

# 6. Frequently Asked Questions (FAQ)

↑ Top

**Q1 Where are my prompts stored? Is my prompt data private and secure?**

Your prompts are stored 100% locally on your machine inside your browser's persistent IndexedDB database. Nothing is transmitted to external servers or cloud analytics. If you run the included optional Python backend ( `server/server.py` ), prompts are saved to an offline SQLite file ( `promptlibrary.db` ).

**Q2 How does the Upfront Variable Pre-fill workflow work?**

Clicking "⚡ **Use Prompt**" opens the Upfront Variable Runner. All variables are presented upfront at the top of the modal. Seeing all variables at once gives you a bird's-eye view of your project flow. As you fill text fields or click choice pills, PromptVault Pro calculates live token consumption ( `Math.ceil(chars / 4)` ). You can copy the compiled prompt or raw template and launch any LLM with 1 click.

**Q3 How do the Edit Category vs. New Category modes work in the Taxonomy Manager?**

In the **Taxonomy & Deduplication Manager**, the top box features an explicit 2-mode switcher. ✏️ **Edit Category Mode** gives you a clean dropdown showing prompt and subcategory counts (e.g. `Executive Presentations (3 prompts, 9 subcats)` ); selecting one pulls all its linked subcategories into view. ➕ **New Category Mode** gives you an empty, auto-focused text field to name a new category under any Domain Pillar without dropdown clutter.

**Q4 How does the Master Subcategory Pool prevent duplicate tags?**

The 100+ Master Subcategory Pool displays all standard tags. Subcategories that are already linked to the currently selected category are highlighted with **solid vibrant accent backgrounds ( `✓ Roadmaps` )**. Unlinked subcategories appear neutral ( `+ Name` ). Clicking any unlinked chip links it instantly; clicking any highlighted chip unlinks it.

**Q5 What happens if I rename or delete a category? Will I lose my prompts?**

No prompts are ever lost! When you click "✏️ **Rename Category**", PromptVault Pro automatically updates the category name across every assigned prompt in the database. If you delete a category, all associated prompts are safely moved to the `General` fallback category.

**Q6 How does Character-by-Character Tag Autocomplete work?**

When editing a prompt, typing in the Tags field triggers a WordPress-style live autocomplete dropdown. As you type each letter (e.g. `s` then `o` ), existing tags refilter instantly with real-time frequency counts ( `seo (12)` ). Use ↑ ↓ arrows and `Enter` to pick an existing tag, or press `Enter`/comma to create a new tag.

**Q7 How do I delete a prompt?**

Open any prompt in the editor modal and click the red "🗑️ **Delete Prompt**" button at the bottom left. A confirmation modal appears to prevent accidental deletions. Upon confirmation, the prompt is removed and all library indexes update immediately.

**Q8 What is the difference between a Simple Prompt and a 5-Section Blueprint?**

A **Simple Prompt** is a single text block with variables, ideal for routine tasks like emails and social posts. A **5-Section Blueprint** partitions prompts into Role Persona, Core Objective, Dynamic Context Inputs, Negative Guardrails, and Output Schema/Rubric for mission-critical enterprise workflows.

**Q9 How can I bulk-parameterize existing prompts that have no variables?**

Open ⇅ **Data** -> **Turnkey LLM Prompt**. Export your 0-variable prompts, paste them into ChatGPT, Claude, or Gemini alongside the included turnkey prompt. The LLM inserts dynamic `{variables}` and choice dropdowns. Re-import the resulting JSON into PromptVault Pro to enrich your prompts instantly!

**Q10** **Can I run PromptVault Pro completely offline?**

Yes! The web application runs 100% offline via browser caching and local IndexedDB. Alternatively, you can run the included Python server (`python3 server/server.py`) for local SQLite persistence with zero pip dependencies.

# 7. Interactive A-Z Prompting Glossary & Subject Index

↑ Top

Click any subject badge on the right to jump directly to the relevant chapter and section in this user guide.

**Author Attribution (Originator Credits)** — Jump to Chapter 3 →

Standardized metadata fields ( `author` and `authorUrl` ) displaying verifiable creator credits on prompt cards.

**Blank Canvas Preset** — Jump to Chapter 5 →

Data Hub action that purges all prompt cards to give users a fresh workspace while preserving all 44 categories and 7 domain pillars.

**Cascade Rename Architecture** — Jump to Chapter 2 →

Automated database synchronization that cascades category title edits across all matching prompt cards to prevent orphaned data.

**Choice Variable (Dropdown / Pill Syntax)** — Jump to Chapter 4 →

Parameter format `{name: Option 1 | Option 2}` rendering as interactive pills with write-in fallbacks.

**Deduplication Shield** — Jump to Chapter 2 →

Interactive taxonomy system utilizing a 100+ master pool with highlighted active chips to prevent fragmented tags.

**Domain Pillars** — Jump to Chapter 1 →

The 7 immutable functional disciplines organizing all knowledge: Automation, Coding, Business, Marketing, Multimedia, Design, and Education.

**Fuzzy Search (⌘K)** — Jump to Chapter 1 →

Client-side search engine filtering titles, descriptions, sections, variables, and tags with zero network latency.

**IndexedDB Architecture** — Jump to Chapter 1 →

Browser-native transactional database engine storing gigabytes of data privately on the client side without 5MB storage caps.

**Modular Blueprint (5-Section Anatomy)** — Jump to Chapter 3 →

Enterprise prompt structure partitioning prompts into Persona, Objective, Dynamic Context Inputs, Negative Guardrails, and Output Rubric.

**Multi-LLM Dispatcher** — Jump to Chapter 4 →

1-click launcher transferring pre-filled prompts directly to ChatGPT, Claude, Gemini, Perplexity, Cursor, or LibreChat.

**Negative Guardrails** — Jump to Chapter 3 →

Explicit anti-hallucination instructions specifying what the AI must NOT do (e.g. no conversational preamble, no speculative citations).

**Smart Ingest (Bracket Converter)** — Jump to Chapter 5 →

In-app parsing engine that detects raw bracketed text like `[topic]` and automatically converts it to `{topic}` .

**Tag Autocomplete** Jump to Chapter 3 →

WordPress-style live tag suggestion dropdown filtering character-by-character with real-time frequency indicators.

**Token Telemetry** Jump to Chapter 4 →

Real-time estimation of prompt token consumption calculated dynamically via `Math.ceil(characters / 4)`.

**Upfront Variable Pre-fill** Jump to Chapter 4 →

Interactive runner interface allowing users to fill in all parameters before copying or dispatching to an LLM.